**Newborough Warren**

Interactive Web Tools

# Technical Note

*Model Equations · Data Architecture · Rendering*

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# Contents

## Introduction

This technical note documents the model equations, data architecture, and rendering implementation for the three interactive web tools produced by the Newborough Warren hydrology research project (Hollingham 2026). Each tool is a self-contained HTML file with all data, model coefficients, and map geometry embedded at build time. The note is organised in three parts.

**Part A covers the Groundwater Flooding Forecaster,** describing the state-space model recurrence, block transfer functions, and P_flood linear form that drive the per-well flood-risk assessments.

**Part B covers the Hydrological Scenario Viewer**, detailing the equilibrium perturbation equation, scenario parameter sets, and the IDW spatial interpolation used to generate the map surface.

**Part C covers the Seasonal Extremes Scatter**, which plots each well's mean summer minimum against its mean winter maximum relative to the Curreli et al. (2013) eco-hydrological thresholds. For guidance on operating the tools, see the companion User Manual.

## Part A

# Groundwater Flooding Forecaster

## A1. Overview

The Groundwater Flooding Forecaster (forecaster.html) is a single-page application built as a standalone HTML file with embedded JavaScript, CSS, and a JSON data bundle. It was generated by Script 11b (build_forecaster_html) from the NRG analysis pipeline. All model coefficients, well metadata, map geometry, and climatological baselines are baked into a single DATA constant at build time.

## A2. Data Bundle Structure

| Key | Contents |
|---|---|
| cluster_coeffs | Per-cluster SSM coefficients (b1, b2, b3), peak month, P_flood linear-form slope A and intercept B, climatological horizon rainfall, and horizon length. Five clusters: C1 (Lake Edge), C2 (Dune), C3 (Western Residual), C4 (Main Forest), C5 (Coastal Forest). |
| block_tf | Seasonal transfer-function coefficients (b1, b2, c, $R^2$) for each geographic block, split into winter and summer sub-models. |
| P_clim / PET_clim | Monthly climatological precipitation and PET (mm), keyed 1–12, from RAF Valley 2005–2026 means. |
| winter_climatology_mm | Mean Oct–Mar cumulative rainfall (514.6 mm), the denominator for λ. |
| wells | Array of 69 well objects with coordinates, elevation, cluster, default depths, and per-well SSM/P_flood coefficients where available. |
| base_layer | Map extent, Base64 hillshade PNG, and KML feature polylines with styling. |

## A3. Model Equations

### A3.1 Monthly state-space model

The core recurrence relation steps groundwater head forward one month:

$$h(t) = \alpha \cdot h(t-1) + b_1 \cdot \lambda \cdot P\_clim(t) - b_2 \cdot PET\_clim(t)$$

where $\alpha = 1 - b_3$, h is head relative to ground (negative = below), P_clim and PET_clim are monthly means, and λ is the rainfall multiplier. Coefficients are available at both cluster and per-well resolution.

### A3.2 Block transfer functions

At the block level, seasonal transfer functions provide a simpler regression:

$$\text{Winter peak:} \quad h\_peak = b_1 \cdot P\_winter + b_2 \cdot h\_min(summer) + c$$

$$\text{Summer min:} \quad h\_min = b_1 \cdot P\_summer + b_2 \cdot h\_max(winter) + c$$

$R^2$ values range from 0.40 (Lake Edge winter) to 0.91 (Forest summer). When $R^2 < 0.50$ the forecast card displays a low-explanatory-power caveat.

### A3.3 P_flood linear form

Cumulative rainfall to bring groundwater to the surface:

$$P\_flood\ (mm) = A \cdot d + B$$

where d is depth below ground (m, positive), A is slope (mm/m), and B is intercept (mm). The result is normalised as λ = P_flood / P_clim_total.

## A4. Cluster and Block Mapping

| Cluster | Label | Block TF | Peak | Horizon |
|---|---|---|---|---|
| C1 | Lake Edge | Lake_Edge | Feb | 5 mo |
| C2 | Dune | Eastern_Block | Feb | 5 mo |
| C3 | Western Residual | Western_Block | Mar | 6 mo |
| C4 | Main Forest | Forest | Mar | 6 mo |
| C5 | Coastal Forest | Coastal_Forest | Mar | 6 mo |

## A5. Live Met Office Integration

On initialisation, loadLiveData() fetches RAF Valley station data. The parser handles estimated values (asterisk-suffixed) and missing data (“---”). buildLiveBanner() computes Oct–Mar cumulative totals for current and prior hydrological winters, deriving λ = observed / climatological. The display adapts by calendar month (Oct–Dec emphasises the previous winter; Jan–Apr the current; May–Sep the most recently completed).

If the fetch fails, a manual fallback modal accepts pasted valleydata.txt and parses it identically.

## A6. Map Rendering

The map uses a fixed SVG viewBox (1000 × 900) with OSGB coordinates (E 240100–243900, N 362100–365900). Rendering layers: (1) Base64 hillshade PNG at 85% opacity, (2) KML polylines/polygons, (3) well dots coloured by P_flood category, re-rendered on every slider change.

## A7. Build Provenance

Script 11b assembles SSM coefficients (Script 03), block transfer functions (Tables 8/9), cluster metadata, hillshade, KML features, and RAF Valley climatology into a single .html file. Google Fonts (Libre Baskerville, Source Sans 3, JetBrains Mono) are the only external resource; the tool degrades gracefully without them.

Citation: Hollingham, M. (2026). Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales.

## Part B

# Hydrological Scenario Viewer

## B1. Overview

The Scenario Viewer (scenario_viewer.html) is generated by Script 19 (19_spatial_groundwater.py, v2.4.2). All computation is client-side JavaScript; no server requests are made after the initial page load.

## B2. Physical Constants

| Parameter | Value and source |
|---|---|
| Hydraulic conductivity K | 6 m/day (Betson et al. 2002) |
| Forest interception (Corsican pine) | 24% (Freeman 2008) |
| Broadleaf interception (deciduous) | 15% annual mean (Komatsu et al. 2011) |
| Sy floor (C1 / C2–C5) | 6% / 12% |
| Ridge mask threshold | 1.0 m |
| Excluded wells | ceh12 (bedrock), ceh15 (forest slack edge) |

## B3. Δh Model Equation

For each well, baseline equilibrium net recharge is:

$$\mathrm{net}_0 = b_1 \cdot P\_eff_0 - b_2 \cdot PET_0 - b_3 \cdot |h|$$

Under a scenario:

$$\mathrm{net\_sc} = b_1 \cdot P\_eff\_sc - b_2\_sc \cdot PET\_sc - b_3 \cdot |h|$$

where P_eff_sc = P × sP × (1 − I_scenario) for forest wells, PET_sc = PET × sPET, and $b_2$_sc = $b_2$ × $sB_2$. The per-well Δh = net_sc − $\mathrm{net}_0$. For the annual season, Δh is the unweighted mean of winter and summer values. The $b_3$ drainage term cancels in the difference.

## B4. Scenario Parameter Sets

| Scenario | sP_w | sP_s | sPET_w | sPET_s | I_c4/c5 | sB2 |
|---|---|---|---|---|---|---|
| Baseline | 1.00 | 1.00 | 1.00 | 1.00 | 0.24 | 1.00 |
| UKCP18 2050s | 1.10 | 0.85 | 1.05 | 1.20 | 0.24 | 1.00 |
| UKCP18 2080s | 1.20 | 0.70 | 1.10 | 1.35 | 0.24 | 1.00 |
| Clearfell | 1.00 | 1.00 | 1.00 | 1.00 | 0.00 | 1.108 |
| Broadleaf | 1.00 | 1.00 | 1.00 | 1.00 | 0.15 | 1.00 |
| Thinning | 1.00 | 1.00 | 1.00 | 1.00 | 0.12 | 1.054 |

The clearfell sB2 multiplier (×1.108) is BACI-corrected: Edge-tier mean b2 ratio minus Climate Control mean ratio plus 1.0 (= 1.1011 − 0.9931 + 1.0), subtracting background climate drift from the felling signal. The thinning multiplier (×1.054) is the half-perturbation. Both values are loaded dynamically from Script 10e output at viewer generation time via clearfell_common.load_clearfell_b2_multiplier().

## B5. Spatial Interpolation

The map surface uses inverse-distance-weighted (IDW) interpolation with power 1 and k = 8 nearest neighbours. A minimum distance floor of 10 m prevents singularities; points within 5 m of a well return the well's exact value. Interpolation is constrained to the site boundary polygon.

In depth mode, a DEM grid (160 × 110 cells) provides the ground surface. Depth = DEM elevation minus IDW head. Ridge masking suppresses rendering where DEM exceeds the IDW-interpolated DEM surface by > 1.0 m.

## B6. Data Bundle

- WELLS: well objects with name, cluster, coordinates, seasonal heads, Sy, SSM coefficients, DEM ground elevation.
- POLYS: KML polygons (site boundary, forest, clearfell, broadleaf, lake).
- CLIMATE: seasonal baselines (P/PET), cluster heads, cluster betas, monthly climatology.
- DEM_GRID: downsampled elevation grid for ridge masking and depth mode.
- HILLSHADE: Base64 RGBA PNG (1100 × 750 px), site-masked, 32 grey levels.

## B7. Viewer Extent

Fixed at E 240200–243700, N 362400–364800 (OSGB36), covering all 69 wells. Canvas default 640 × 440 px, resizable. A parallel CSV (19_scenario_summary.csv) mirrors the viewer's calculations for the manuscript.

## B8. Build Provenance

Script 19 assembles data from Scripts 00, 01, 03, 17, 18, and site GIS layers.

Citation: Hollingham, M. (2026). Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales.

Part C

# Seasonal Extremes Scatter

## C1. Overview

The seasonal extremes scatter (14_seasonal_extremes_scatter.html) is generated by Script 14 (14_climate_projections.py, v1.1.0). It reads per-well summary statistics from the well network table (00_well_network_table.csv) and cluster assignments from 02_cluster_stats.csv.

## C2. Data Sources

- Mean_Summer_Min_m: mean of annual summer (Apr–Sep) minimum water-table depths, 2005–2026, per well. Values in metres relative to pipe top (negative = below).
- Mean_Winter_Max_m: mean of annual winter (Oct–Mar) maximum depths, computed identically.
- Cluster: from 02_cluster_stats.csv. Unmatched wells labelled "UNKNOWN".
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## C3. Threshold Definitions

Ecological thresholds from Curreli et al. (2013), imported from utils/config.py:

| Constant | Value (m) | Meaning |
|---|---|---|
| SD15b (summer) | 0.61 | Summer minimum viability limit for wet dune slack. |
| SD16 (summer) | 0.98 | Summer minimum viability limit for dry dune slack. |
| SD15b_WINTER | 0.10 | Winter maximum flooding threshold for wet slack. |
| SD16_WINTER | 0.25 | Winter maximum flooding threshold for dry slack. |

## C4. Chart Implementation

Uses Chart.js 4.4.1 from cdnjs. Well data serialised as JSON point objects. A custom plugin (thresholdPlugin) draws the four threshold lines after the scatter data. The search mechanism rebuilds datasets with modified colours and radii: matched well in orange (#ff6600) at radius 13, others at 33% opacity.

## C5. Cluster Styling

Colours, labels, and markers defined in utils/config.py (CLUSTER_COLOURS, CLUSTER_LABELS, CLUSTER_MARKERS) as the single source of truth. Five clusters under the k = 5 partition: C1 Lake Edge, C2 Dune, C3 Western Residual, C4 Main Forest, C5 Coastal Forest.

## C6. Build Provenance

Script 14 also produces static PNG figures (summer trajectory, winter flooding, stacked two-panel), a summer trend CSV, annual extremes CSV, and winter exceedance summary. The scatter HTML is an additional interactive output complementing these.

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